



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SARAL PETROLEUM

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

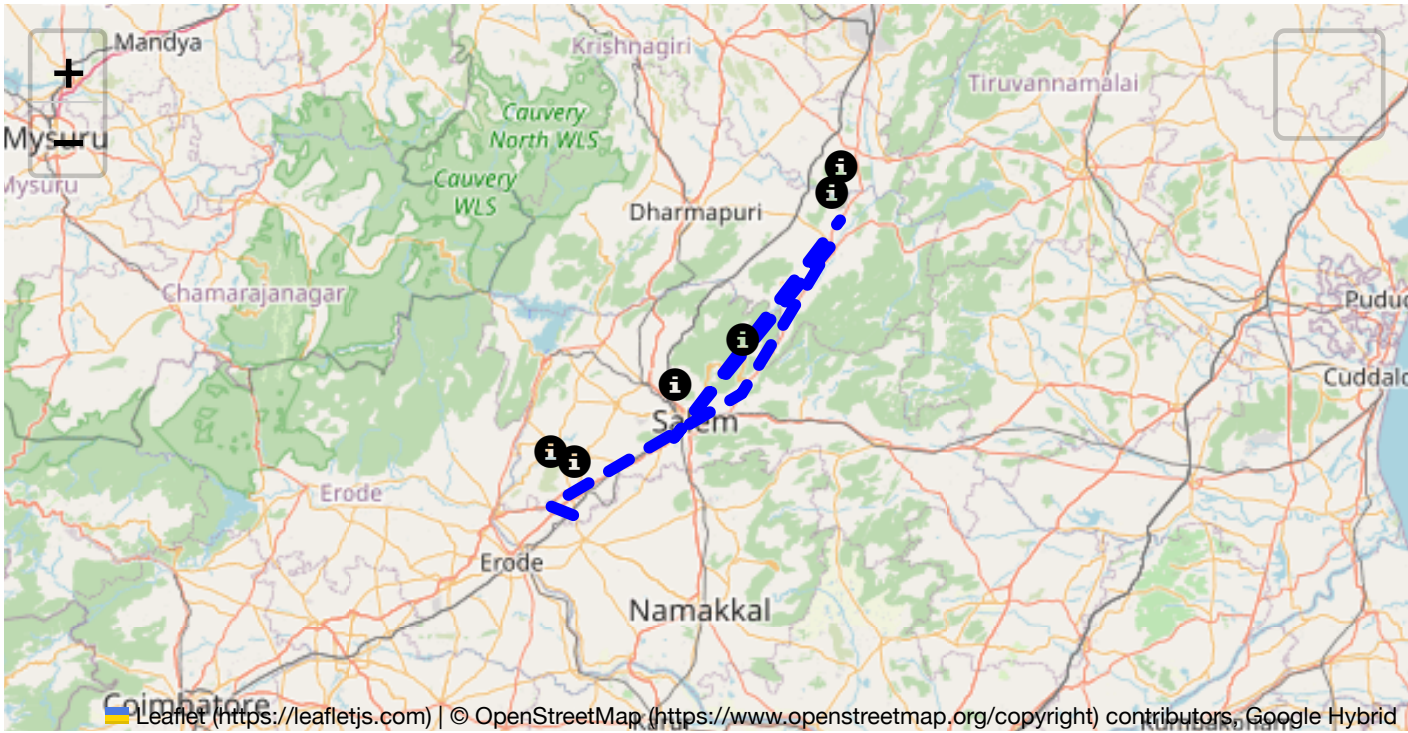
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

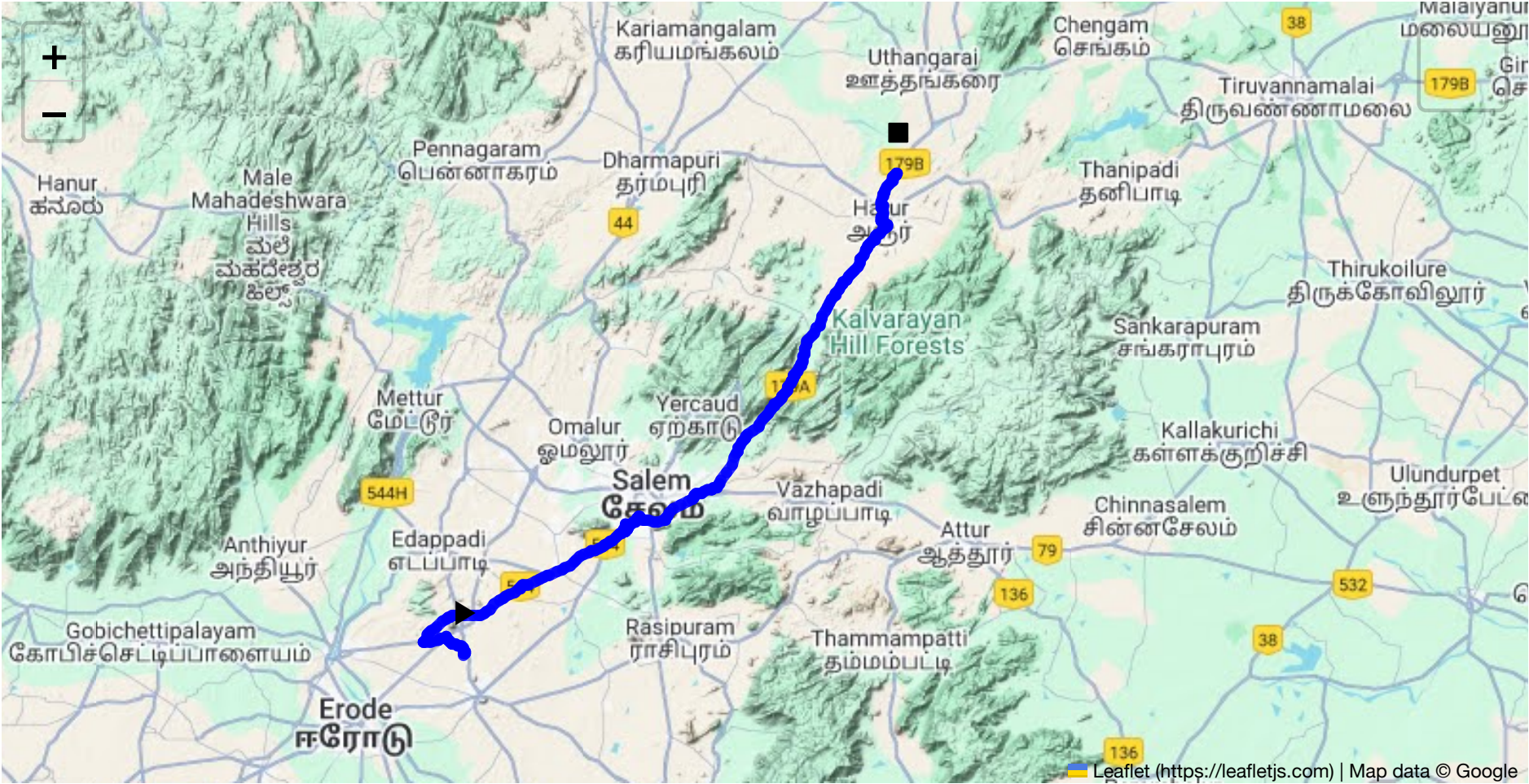
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 127.47 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours

Start: (11.4381, 77.8734)
End: (12.115814, 78.496888)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 4F8W+CV2, H. Eachambadi, via key points in Padaiveedu, Sukkampatti, Govindasamy Nagar, and the Nagapattinam-Coimbatore-Gundlupet Highway, covers approximately 127.47 kilometers and is traversed primarily on state and national highways. The journey involves navigating local arterial roads and bypass routes, indicative of mixed traffic conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate, with hot summers, a significant monsoon season from October to December, and relatively dry conditions in winter.

- **Summer (March to June):** Heat can lead to road surface melting, causing slippery conditions.
- **Monsoon (October to December):** Heavy rainfall may result in water-logged roads, reduced visibility, and disrupted traffic flow.
- Drivers should be prepared for sudden localized storms during the monsoon and occasional fog in the early mornings during winter.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, peak traffic occurs between 8-10 AM and 6-8 PM.
- **Congestion-Prone Areas:** Major towns like Salem, known for educational institutions and industrial nodes, might face notable congestion.
- Specific bottlenecks may occur at intersections on the Salem Bypass and near industrial zones.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Main highways are generally well-maintained, but local roads may vary, with potential for potholes and uneven surfaces.
- **Infrastructure:** Mixed conditions with sections having advanced infrastructure like bypasses and rest stops, while other stretches may lack proper signage and lighting.

5. Suggestions for Alternative Routes for Emergencies

- Alternative routes may involve:
 - Using NH44, a broader national highway parallel to parts of the described route.
 - Diversions through less congested rural roads for avoiding peak traffic but verifying road conditions and accessibility are crucial.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Required for specific hazardous materials, adhering to government regulations.
- **Timings:** Restrictions may apply on hazardous material transportation during peak traffic times in urban areas.

7. Overview of Historical Incidents

- Information on historical incidents along this route suggests occasional truck overturning or road accidents, typically due to over-speeding or poor visibility during adverse weather. There are no major incident records involving hazardous materials recently.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Portions of the route may pass near ecologically sensitive zones; compliance with environmental regulations is crucial.
- **Noise and Emissions:** Adherence to emission norms is essential, especially near residential and wildlife areas.

9. Analysis of Communication Coverage

- **Coverage Quality:** Major carrier networks typically provide robust coverage along highways, including 3G and 4G. Expect potential dead zones in rural or forested areas.
- **Dead Zones:** Likely in more remote sections between major urban centers like Salem and Harur.

10. Estimated Emergency Response Times

- **Urban Centers (e.g., Salem):** Quick response times averaging 15-30 minutes.
- **Rural Areas:** Extended to 45-60+ minutes due to lower density of emergency services.

12. Overall Summary of Risk Assessment

- The route, though generally navigable with good infrastructure over significant stretches, has variable conditions affected by seasonal weather, road quality differences, and potential congestion. It's

crucial for drivers to adapt to changing conditions, remain aware of legal compliance for hazardous materials, and have contingency plans for communication or route diversions. Monitoring weather forecasts and traffic reports, alongside maintaining vehicle conditions, can largely mitigate risk and ensure safer transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	U-Turn	High	11.6180669, 78.1109048	10 KM/Hr
2	U-Turn	High	11.6128439, 78.1059968	10 KM/Hr
3	U-Turn	High	12.1178163, 78.4982242	10 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	High	11.44900, 77.87406	15 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45781, 77.85203	30 KM/Hr
9	Turn	High	11.46307, 77.84941	15 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45605, 77.81683	15 KM/Hr
12	Turn	Medium	11.53573, 77.95794	30 KM/Hr
13	Turn	Medium	11.62100, 78.10723	30 KM/Hr
14	Blind Spot	Blind Spot	11.62130, 78.10776	10 KM/Hr
15	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
16	Turn	Medium	11.61235, 78.10495	30 KM/Hr
17	Blind Spot	Blind Spot	11.61807, 78.11090	10 KM/Hr
18	Blind Spot	Blind Spot	11.61286, 78.10611	10 KM/Hr
19	Blind Spot	Blind Spot	11.63306, 78.12722	10 KM/Hr
20	Turn	Medium	11.62821, 78.13348	30 KM/Hr
21	Turn	Medium	11.62851, 78.16509	30 KM/Hr
22	Turn	Medium	11.65782, 78.20430	30 KM/Hr
23	Turn	High	11.66387, 78.20665	15 KM/Hr
24	Turn	High	11.67131, 78.23138	15 KM/Hr
25	Turn	High	11.67195, 78.23146	15 KM/Hr
26	Turn	Medium	11.67433, 78.23905	30 KM/Hr
27	Turn	Medium	11.80416, 78.33319	30 KM/Hr
28	Turn	Medium	11.81219, 78.34077	30 KM/Hr
29	Turn	Medium	11.83472, 78.35285	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	Medium	11.99572, 78.44559	30 KM/Hr
31	Turn	Medium	12.01548, 78.45650	30 KM/Hr
32	Turn	Medium	12.04309, 78.48361	30 KM/Hr
33	Turn	Medium	12.04429, 78.48380	30 KM/Hr
34	Turn	Medium	12.04490, 78.48355	30 KM/Hr
35	Turn	High	12.04984, 78.47480	15 KM/Hr
36	Turn	Medium	12.06014, 78.47787	30 KM/Hr
37	Turn	Medium	12.06962, 78.47655	30 KM/Hr
38	Turn	Medium	12.07207, 78.47727	30 KM/Hr
39	Turn	Medium	12.07536, 78.47607	30 KM/Hr
40	Turn	Medium	12.11562, 78.49617	30 KM/Hr
41	Blind Spot	Blind Spot	12.11778, 78.49830	10 KM/Hr
42	Turn	Medium	12.11665, 78.49779	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
20	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
21	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
22	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
24	hospital	Dr.Thiru Neuro Multi Speciality	11.6623376, 78.203353	30 km/h	Medium
26	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
27	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
28	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
36	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium
37	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
23	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
29	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
30	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
31	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
32	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
33	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
34	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium
35	school	Government Girls Hr Sec School	12.0488709, 78.4762447	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.6180669, 78.1109048



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.6128439, 78.1059968



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 12.1178163, 78.4982242



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62100, 78.10723



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61235, 78.10495



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61807, 78.11090



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61286, 78.10611



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63306, 78.12722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62821, 78.13348



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62851, 78.16509



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65782, 78.20430



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66387, 78.20665



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Speed Limit: 15 KM/Hr
Coordinates: 11.67131, 78.23138



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Speed Limit: 15 KM/Hr
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Speed Limit: 30 KM/Hr
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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04429, 78.48380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04490, 78.48355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04984, 78.47480



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06014, 78.47787



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Speed Limit: 30 KM/Hr
Coordinates: 12.06962, 78.47655



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Speed Limit: 30 KM/Hr
Coordinates: 12.07207, 78.47727



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07536, 78.47607



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11562, 78.49617



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.11778, 78.49830



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11665, 78.49779

